

Mitsue Project — Earned Value Management Plan

Performance Baseline · Cost Control · Forecast Framework

Phases 0–3 · April 2026 – September 2028

1. Purpose and Scope

This EVM Plan establishes the **Performance Measurement Baseline (PMB)** for the Mitsue Sustainable Energy & AI Data Center project across its four build phases (Phase 0 through Phase 3). It defines how the project will measure schedule and cost performance, report progress to stakeholders, and forecast the cost and date at completion.

The plan covers the **30-month period April 2026 – September 2028**. Phase 4 (Operations & Scale, Month 31+) is not included in the baseline because it is funded differently (operating revenue) and planned in Year 2 once Phase 1 results are confirmed.

Data Date for current baseline: 2026-05-18 (end of Month 2)

2. EVM Fundamentals — Key Metrics

Term	Symbol	Formula	What it means
Planned Value	PV	Time-phased budget	Budgeted cost of work scheduled to date
Earned Value	EV	% complete × BAC	Budgeted cost of work actually performed
Actual Cost	AC	Invoices + payroll	Real money spent to date
Budget at Completion	BAC	Sum of all PV	Total approved budget for the PMB
Schedule Variance	SV	EV – PV	Negative = behind plan
Cost Variance	CV	EV – AC	Negative = over budget
Schedule Performance Index	SPI	EV ÷ PV	< 1.0 = behind schedule
Cost Performance Index	CPI	EV ÷ AC	< 1.0 = over budget
Estimate at Completion	EAC	AC + (BAC–EV)÷CPI	Projected final cost
Estimate to Complete	ETC	EAC – AC	Remaining work cost forecast
Variance at Completion	VAC	BAC – EAC	Projected over/under at end
To-Complete Perf. Index	TCPI	(BAC–EV)÷(BAC–AC)	Efficiency needed to finish on budget

3. Budget Structure

3.1 Summary Budget

Level	Item	Budget (¥M)	% of BAC
Total Project Budget		¥245.0M	—
— Management Reserve (~11%)	Not in PMB	¥25.0M	—
Performance Measurement Baseline (BAC)		¥220.0M	100%
1.0	Project Management & Governance	¥15.0M	6.8%
2.0	Phase 0 — Pre-Foundation	¥0.25M	0.1%
3.0	Phase 1 — Foundation	¥5.5M	2.5%
4.0	Phase 2 — Pilot Design	¥22.25M	10.1%
5.0	Phase 3 — Pilot Build	¥177.0M	80.5%

Management Reserve (¥25M) is held by the Representative Director and is **not part of the PMB**. It may be drawn down only with explicit board approval to address unforeseen scope.

Baseline Rev 1 (May 2026) — Reality-checked against real-world Japan benchmarks for solar PV (¥200–300K/kW), school seismic retrofit (¥200–500K/m²), commercial EV fast-chargers (¥5–6M each), and forestry road works. Phase 3 line items, PM/governance, and contingency uplifted accordingly; BAC raised from ¥168M to ¥220M.

3.2 Work Breakdown Structure (WBS) — Budget Detail

WBS	Element	Budget (¥M)	Phase
1.0	Project Management & Governance		
1.1	Core PM team (Rep. Director + coordinator)	¥8.0M	All
1.2	Legal, accounting, 行政書士 fees	¥4.0M	All

WBS	Element	Budget (¥M)	Phase
1.3	Communications, website, translation	¥3.0M	All
	<i>Sub-total 1.0</i>	<i>¥15.0M</i>	
2.0	Phase 0 — Pre-Foundation		
2.1	Community stakeholder meetings & travel	¥0.10M	P0
2.2	Charter & document preparation	¥0.10M	P0
2.3	Founding team identification	¥0.05M	P0
	<i>Sub-total 2.0</i>	<i>¥0.25M</i>	
3.0	Phase 1 — Foundation		
3.1	一般社団法人 incorporation	¥0.5M	P1
3.2	Forestry feasibility study	¥1.5M	P1
3.3	Energy systems feasibility study	¥1.5M	P1
3.4	Building & site assessment	¥0.8M	P1
3.5	Connectivity assessment	¥0.4M	P1
3.6	Advisory board, bank, accounting setup	¥0.8M	P1
	<i>Sub-total 3.0</i>	<i>¥5.5M</i>	
4.0	Phase 2 — Pilot Design		
4.1	Structural & architectural design	¥2.0M	P2
4.2	Energy systems engineering	¥2.0M	P2
4.3	Data center & IT design	¥1.0M	P2
4.4	EV charging system design	¥1.0M	P2
4.5	Permitting & regulatory (METI, FIT, forestry)	¥3.5M	P2
4.6	Partnership & landowner agreements	¥1.5M	P2
4.7	Grant writing & funding applications	¥2.0M	P2
4.8	Staff hiring & onboarding (2–3 part-time)	¥6.0M	P2
4.9	Vendor pre-qualification	¥1.5M	P2
4.10	Phase 2 contingency	¥1.75M	P2
	<i>Sub-total 4.0</i>	<i>¥22.25M</i>	
5.0	Phase 3 — Pilot Build		
5.1	School building renovation (1 wing)	¥38.0M	P3
5.2	Solar PV installation (~100 kW)	¥22.0M	P3
5.3	Battery storage system	¥12.0M	P3
5.4	EV charging infrastructure (4 stations)	¥15.0M	P3
5.5	Data center fitout (10–20 servers)	¥20.0M	P3
5.6	Forestry operations (5–10 ha harvest + replant)	¥25.0M	P3
5.7	Fiber connectivity upgrade	¥10.0M	P3
5.8	Testing, commissioning, and startup	¥8.0M	P3
5.9	Phase 3 contingency (18%)	¥27.0M	P3
	<i>Sub-total 5.0</i>	<i>¥177.0M</i>	
	TOTAL BAC	¥220.0M	

4. Time-Phased Baseline (S-Curve)

Monthly planned expenditure and cumulative planned value over the 30-month baseline. Spending is front-loaded within phases on studies and design; construction spending peaks in Months 22–27.

Month	Calendar	Phase	Monthly PV (¥M)	Cumulative PV (¥M)	% Complete
M1	Apr 2026	P0	0.05	0.05	0.0%
M2	May 2026	P0	0.10	0.15	0.1%
M2 ← Status Date					
M3	Jun 2026	P0	0.10	0.25	0.1%
M4	Jul 2026	P1	0.20	0.45	0.2%

Month	Calendar	Phase	Monthly PV (¥M)	Cumulative PV (¥M)	% Complete
M5	Aug 2026	P1	0.50	0.95	0.4%
M6	Sep 2026	P1	0.80	1.75	0.8%
M7	Oct 2026	P1	1.20	2.95	1.3%
M8	Nov 2026	P1	1.50	4.45	2.0%
M9	Dec 2026	P1	1.30	5.75	2.6%
M10	Jan 2027	P2	1.50	7.25	3.3%
M11	Feb 2027	P2	2.20	9.45	4.3%
M12	Mar 2027	P2	2.80	12.25	5.6%
M13	Apr 2027	P2	3.00	15.25	6.9%
M14	May 2027	P2	3.00	18.25	8.3%
M15	Jun 2027	P2	3.00	21.25	9.7%
M16	Jul 2027	P2	2.50	23.75	10.8%
M17	Aug 2027	P2	2.50	26.25	11.9%
M18	Sep 2027	P2	2.00	28.25	12.8%
M19	Oct 2027	P3	4.00	32.25	14.7%
M20	Nov 2027	P3	7.00	39.25	17.8%
M21	Dec 2027	P3	11.00	50.25	22.8%
M22	Jan 2028	P3	16.50	66.75	30.3%
M23	Feb 2028	P3	20.50	87.25	39.7%
M24	Mar 2028	P3	20.50	107.75	49.0%
M25	Apr 2028	P3	25.00	132.75	60.3%
M26	May 2028	P3	25.00	157.75	71.7%
M27	Jun 2028	P3	20.50	178.25	81.0%
M28	Jul 2028	P3	20.50	198.75	90.3%
M29	Aug 2028	P3	14.00	212.75	96.7%
M30	Sep 2028	P3	7.25	220.00	100.0%
	BAC		¥220.0M		

S-Curve Shape

The expenditure profile follows a classic **slow-fast-slow** S-curve:

- **M1–M9** (Phases 0–1): Ramp-up — legal, feasibility studies, advisory engagement
- **M10–M18** (Phase 2): Acceleration — engineering design, permitting, grant applications
- **M19–M28** (Phase 3 core): Peak spend — construction, procurement, installation
- **M29–M30** (Phase 3 close): Tail-off — commissioning, punch-list, handover

Figure 1 — Baseline S-Curve (Cumulative Planned Value)

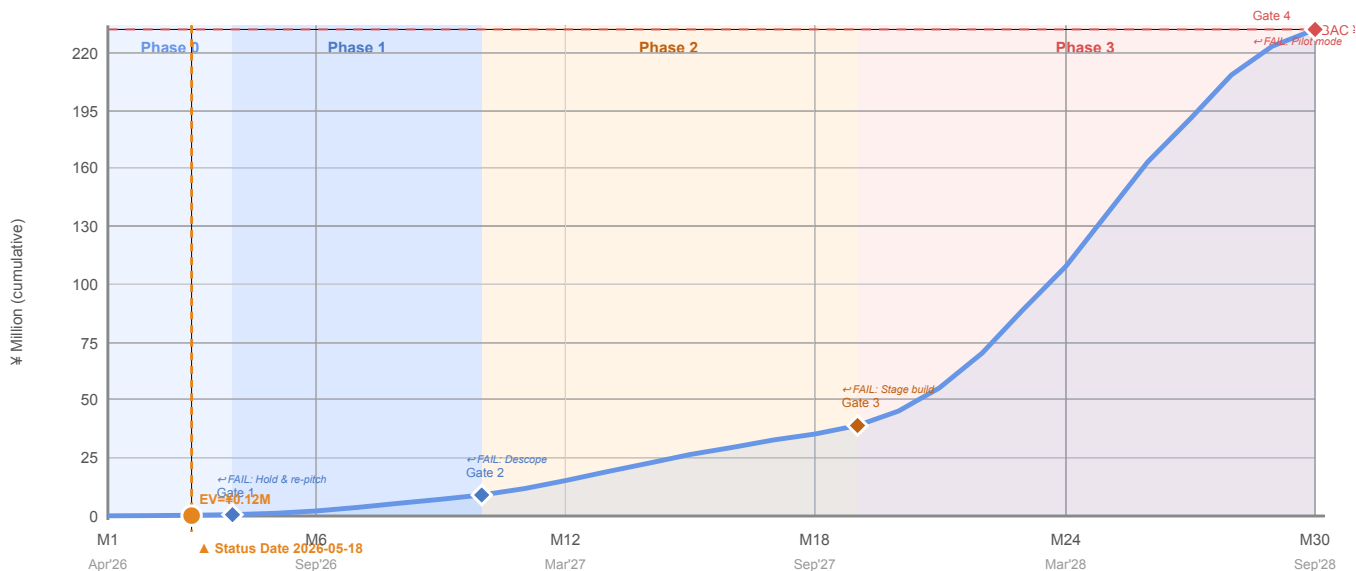
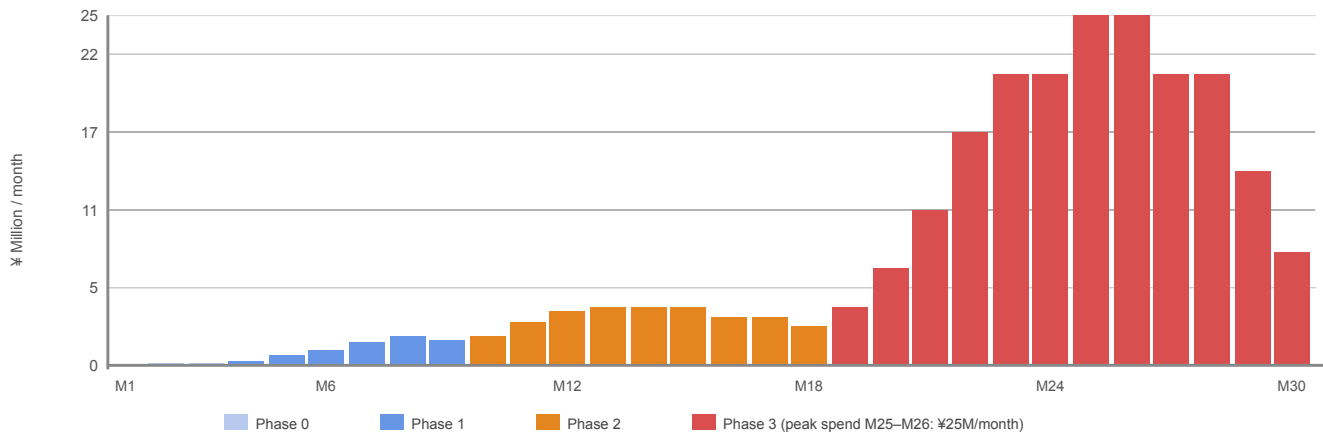


Figure 2 — Monthly Planned Expenditure by Phase



5. Current Performance Status

Data Date: 2026-05-18 (End of Month 2)

5.1 Status Narrative

The project is in **Phase 0 — Pre-Foundation**. Activities in progress include quiet stakeholder conversations with village leadership, preliminary charter drafting, and identifying founding team candidates. The Japanese co-founder search is underway but not yet resolved — this is the single most critical Phase 0 dependency. Expenditure to date is personal travel, document translation, and a preliminary consultation with a Nara-based 行政書士. No major contracts have been signed.

5.2 Performance Metrics — Phase 0

Metric	Value	Notes
PV (Planned Value to date)	¥0.15M	Planned spend through M2
EV (Earned Value to date)	¥0.12M	~80% of planned Phase 0 work done
AC (Actual Cost to date)	¥0.08M	Under plan — mostly personal time
SV (Schedule Variance)	-¥0.03M	Slightly behind: co-founder not yet identified
CV (Cost Variance)	+¥0.04M	Under budget: Phase 0 costs lower than planned
SPI	0.80	80% of planned work achieved on schedule
CPI	1.50	Each ¥1 spent is delivering ¥1.50 of value

Figure 3 — Current EVM Performance Dashboard (Data Date: 2026-05-18)



Note: CPI of 1.50 at Month 2 is partly a Phase 0 artefact — most value creation is through personal time rather than budgeted spend. Do not extrapolate CPI to total project forecast until Phase 1 feasibility contracts are in place.

5.3 Phase 0 Forecast

Metric	Value
Phase 0 BAC	¥0.25M
EAC (Phase 0) = AC + (BAC-EV)÷CPI	¥0.16M
ETC (Phase 0) = EAC - AC	¥0.08M
VAC (Phase 0) = BAC - EAC	+¥0.09M under budget
TCPI	0.87 — need only 0.87 efficiency to finish Phase 0 within budget

6. Phase-by-Phase Forecast (Baseline Projections)

These projections assume execution at the planned efficiency levels. They will be revised monthly from Phase 1 onwards when contract values and feasibility study costs are confirmed.

6.1 Phase 0 — Pre-Foundation

Item	Value
Duration	M1–M3 (Apr–Jun 2026)
BAC	¥0.25M
Planned end date	2026-06-30
Key gate	Founding team verbal commitments; village leadership informed
Current forecast end	On track for Jun 2026
Current forecast cost	¥0.16–0.22M

6.2 Phase 1 — Foundation

Item	Value
Duration	M4–M9 (Jul–Dec 2026)
BAC	¥5.5M
Planned end date	2026-12-31
Key gate	¥3–8M secured; legal entity registered; feasibility studies complete
Critical path	Forestry + Energy feasibility studies (procurement Q3 2026)
Primary risk	Government grant approvals often take 2–3 months longer than planned
Contingency plan	Reduce feasibility scope slightly; delay P2 start by 1–2 months

6.3 Phase 2 — Pilot Design

Item	Value
Duration	M10–M18 (Jan–Sep 2027)
BAC	¥22.25M
Planned end date	2027-09-30
Key gate	¥30–50M secured; detailed engineering complete; key permits in hand
Critical path	METI permitting + FIT registration (can take 6+ months)
Primary risk	Permit delays push Phase 3 start past Oct 2027
Contingency plan	Stage Phase 3 procurement independently of permit approvals for non-permit items

6.4 Phase 3 — Pilot Build

Item	Value
Duration	M19–M30 (Oct 2027–Sep 2028)
BAC	¥177M
Planned end date	2028-09-30
Key gate	Revenue operational; Phase 4 plan confirmed
Critical path	Building renovation → solar/battery install → EV charging → commissioning
Primary risk	Building structural findings requiring scope increase (currently low probability, high impact)
Contingency plan	Phase 3 WBS 5.9 includes ¥27M contingency (18%); Management Reserve of ¥25M backstop

7. Three-Scenario Forecast

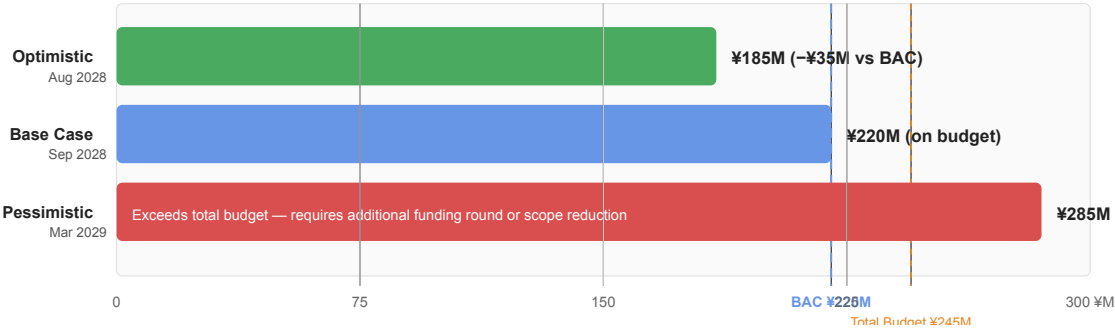
For a project with this level of regulatory and geographic complexity, planning in three scenarios is more honest than a single-point estimate.

Scenario	Description	EAC	End Date
Optimistic	All grants awarded on first application; building in good condition; no permit delays; lower-bound vendor quotes	¥185M	Aug 2028
Base Case	One major permit delay (+2 months); one grant deferred to Year 2; CPI ≈ 1.0; mid-range vendor quotes	¥220M	Sep 2028

Scenario	Description	EAC	End Date
Pessimistic	Two permit delays; building remediation required; government grant shortfall — draws on MR and beyond	¥285M	Mar 2029

The scenario band is intentionally wider than the Rev 0 baseline. Phase 1 feasibility studies (M9) will narrow the range substantially once school structural condition, vendor quotes, and grant awards are confirmed.

Figure 4 — Three-Scenario EAC Comparison



The pessimistic scenario at ¥285M **exceeds the total approved budget of ¥245M by ¥40M** and would trigger a Gate 3 funding decision: either (a) raise an additional funding round, (b) reduce Phase 3 scope (e.g., defer EV chargers or fiber upgrade), or (c) extend the schedule and stage construction. Absolute failure scenarios (community resistance, no funding at Gate 1) are still managed at the funding gate level, not the EVM level.

8. Performance Thresholds and Escalation

EVM metrics trigger review actions at the following levels. Thresholds are intentionally tighter in early phases when course-correction is cheapest.

Threshold	SPI or CPI	Action
Green	0.95 – 1.10	Normal reporting; no action required
Yellow	0.85 – 0.94	Representative Director notified; recovery plan drafted within 2 weeks
Orange	0.75 – 0.84	Board briefing required; recovery plan presented with options
Red	< 0.75	Formal board review; consider phase rescoping or Management Reserve draw

Schedule threshold override: If SPI < 0.90 at any funding gate checkpoint, the gate decision is automatically deferred pending a recovery assessment, regardless of CPI.

9. EVM Reporting Cadence

Frequency	Report	Recipients	Content
Monthly	EVM Status Update	Board, key funders	PV/EV/AC table; SPI/CPI; SV/CV; EAC update; issues
At each Gate	Gate Review Package	Board + major funders	Full EVM review; scenario forecast; go/no-go recommendation
Quarterly	Stakeholder Summary	Village government, advisors	High-level cost and schedule status; milestone progress
Annual	Year-End Review	All stakeholders	Full financial report; EVM audit; next-year baseline

Monthly reports are generated from OpenProject time tracking + invoicing. The first formal monthly EVM report is due **2026-06-30** (end of Phase 0).

10. Assumptions and Constraints

Assumptions

- Project start date: 2026-04-01 (Month 1)
- All phase durations are calendar months, including Japanese holidays
- Feasibility studies are awarded to single vendors per study; no multi-vendor splits
- Phase 3 construction prices are based on 2026 Nara Prefecture rural construction indices, with a 15–25% rural mobilisation premium applied to commercial benchmarks for solar PV, EV charging, and seismic retrofit work
- Exchange rate assumptions (for Dutch/international corporate partnerships): ¥150/EUR
- Forestry operations are weather-dependent; M19–M23 baseline avoids typhoon season

Constraints

- BAC of ¥220M is a **ceiling**; Phase 3 cannot proceed without confirmed funding at Gate 3
- Management Reserve requires board approval for any drawdown
- EVM data must be reported in JPY; foreign currency transactions converted at transaction-date rate
- The Phase 1 feasibility studies are the **single largest driver of overall project accuracy** — all EAC figures before M9 carry high uncertainty (±40%)

11. Limitations and Honest Caveats

This is an **early-stage EVM plan** written at Month 2 of a 30-month project. The following limitations apply:

1. **CPI and SPI at Month 2 are not statistically reliable.** With only ¥0.08M of actual cost data, the performance indices reflect Phase 0 characteristics (personal time, low hard costs) rather than the project overall. Trust these numbers only after ¥2–3M of Phase 1 costs are confirmed.
2. **The Phase 3 budget range is wide.** The ¥120–290M range from the implementation plan reflects genuine uncertainty about building condition (¥30–100M school renovation range alone), equipment procurement conditions, and grid connection costs. The EVM baseline uses ¥177M for WBS 5.0; the ¥27M Phase 3 contingency (18%) and ¥25M Management Reserve exist precisely to absorb this range.
3. **Funding gate gating is the primary control mechanism.** EVM monitors cost and schedule efficiency within a phase; the funding gates control whether the next phase begins at all. These two controls are complementary, not redundant.
4. **Phase 1 feasibility studies will substantially revise this baseline.** After M9, expect a formal baseline revision (Rev 2). The revised Phase 2 and Phase 3 budgets will be anchored to real survey data — particularly the school structural assessment and vendor quotes — not Japan-market benchmarks.

12. Baseline Revision Policy

The PMB may be formally revised:

- After completion of Phase 1 feasibility studies (mandatory revision at M9)
- At any funding gate when confirmed funding differs from planned by more than ±20%
- When board-approved scope changes are incorporated
- When Management Reserve is drawn down

Each baseline revision is documented with: old baseline, new baseline, reason, approving authority, and date.

Revisions are numbered:

- **Rev 0** (v1.1, Apr 2026) — Original BAC ¥168M, drafted from planning estimates
- **Rev 1** (v2.0, May 2026) — Current document. BAC ¥220M, reality-checked against real-world Japan benchmarks (solar, EV chargers, school seismic retrofit, forestry). Reason: pre-emptive correction in advance of Phase 1 feasibility studies.
- **Rev 2** (planned M9, Dec 2026) — Post-Phase 1 revision anchored to feasibility study results and vendor quotes.

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